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Overestimated global dryland expansion with substantial
increases in vegetation productivity under climate warmingZiwei Liu[✉], Taihua Wang[✉] and Hanbo Yang^{*}State Key Laboratory of Hydro-Science and Engineering, Department of Hydraulic Engineering, Tsinghua University, Beijing,
People's Republic of China^{*} Author to whom any correspondence should be addressed.E-mail: yanghanbo@tsinghua.edu.cn**Keywords:** dryland expansion, potential evapotranspiration, CO₂ fertilization, stomatal behavior, dryland vegetation productivity
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Abstract

Drylands are serviced as an essential component of the earth's ecosystem. The potential changes in dryland areas are of great importance to the environment, but various debates remain as to whether and to what extent drylands are expected to expand. Here we employ a physically-based potential evapotranspiration (E_p) model accounting for vegetation response to climate change to quantify potential changes in dryland areas, on the basis of a commonly used indicator, aridity index (multiyear mean E_p over precipitation). Results show that by the end of this century, drylands will expand slightly by $\sim 5\%$, while vegetation productivity will increase by $\sim 50\%$. Elevated CO₂ slows down the increase rate of E_p that impedes the expansion of drylands, but greatly promotes vegetation growth with increases in both leaf assimilation and canopy foliage. These findings improve our understanding of the potential changes in dryland and their ecological impacts in a warmer climate.

1. Introduction

Drylands generally refer to the regions with water stress, covering $\sim 40\%$ of the land surface, and are essential for global environments and sustainability (Lian *et al* 2021). With climate warming, whether dryland will expand remains a matter of debate. Some literature, using potential evapotranspiration (E_p)-based indicators, such as aridity index (AI, the ratio of E_p to precipitation) and/or Palmer drought severity index (Palmer (1965)), reported that drylands will significantly expand with rising temperature (Dai 2013, Feng and Fu 2013, Huang *et al* 2016, Park *et al* 2018, Yao *et al* 2020). Nevertheless, this conflicts with the findings on the basis of other indicators, such as surface runoff and/or eco-hydrological indices (Sheffield *et al* 2012, Yang *et al* 2018, Berg and McColl 2021), which show little or no change in dryland areas. The estimated rapid expansion in dryland areas generally comes from the dramatic increases in E_p , calculated by the Penman–Monteith (Penman 1948, Allen *et al* 1998) and/or temperature-based (Thornthwaite 1948) models. However, the responses

of vegetation to changing environments, especially CO₂ fertilization, are often neglected in those traditional E_p models (Roderick *et al* 2015, Milly and Dunne 2016), possibly resulting in an overestimated arid future and causing conflicts between atmospheric and hydrologic indicators (Milly and Dunne 2016, Swann *et al* 2016, Yang *et al* 2019, Berg and McColl 2021, Lian *et al* 2021).

Another focus of concern is the potential changes in drylands vegetation productivity, represented by gross primary productivity (GPP). With satellite observations, the positive effects of elevated CO₂ are found to overwhelm the negative effects of rising temperature (shown as vapor pressure deficit), promoting vegetation growth in the past decades (Donohue *et al* 2013, Ukkola *et al* 2016, He *et al* 2019, Zhang *et al* 2020, Manzoni *et al* 2022). Nevertheless, a recent study claimed that the imbalance of GPP between expanded and historical drylands may lead to decreases in GPP per unit dryland area (Yao *et al* 2020), which seems to run counter to the expectation. Specifically, the increases in total ecosystem-level GPP come from two components, i.e. changes in

leaf-level assimilation rate (A) and canopy structure (represented by leaf area index, LAI), and how these two components contribute to GPP changes remains unclear yet. Besides, the quantification of the positive role of elevated CO_2 and the negative impacts of rising atmospheric vapor deficit (VPD) on GPP changes also remains to be explored (Donohue *et al* 2017, Manzoni *et al* 2022).

Motivated by the above issues, based on Coupled Model Inter-comparison Project Phase 5 (CMIP5) climate models (Taylor *et al* 2012), we aim to answer two scientific questions: (a) with a physically-based and vegetation-accounted E_p model, will the drylands expand or not, and what will be the magnitude of change? (b) with the re-identified dryland area changes, how does the vegetation productivity indicated by GPP change, and what are the relative contributions of two opposite effects from elevated CO_2 and rising VPD? To answer the first question, we develop a new E_p equation based on an optimal surface resistance model that quantifies the vegetation response to climate change. For the second question, we first analyze the direct GPP output from the CMIP models and then use a bulk model to decompose the respective contributions of A and LAI on GPP changes. A minimalistic model is further employed to investigate the role of CO_2 and VPD on GPP changes across the global drylands in a changing climate.

2. Method

2.1. E_p model

The original Penman–Monteith (PM) equation encapsulates the influencing factors for E_p estimation as (Penman 1948, Monteith 1965):

$$E_{p_PM} = \frac{1}{\lambda} \frac{\Delta R_n + \rho_a C_p \text{VPD}/r_a}{\Delta + \gamma(1 + r_s/r_a)} \quad (1)$$

where λ (J kg^{-1}) is the latent heat of water vaporization, Δ (kPa K^{-1}) is the slope of the saturated vapor pressure versus temperature curve, R_n (W m^{-2}) is the available radiation, γ is the psychrometric constant, ρ_a (kg m^{-3}) is the air density, C_p ($1010 \text{ J kg}^{-1} \text{ K}^{-1}$) is the specific heat of the air at constant pressure, r_a (s m^{-1}) is the aerodynamic resistance, VPD is the vapor pressure deficit calculated by air temperature and humidity (Murray 1967), and r_s (s m^{-1}) is the surface resistance.

To determine r_a and r_s , a kind of reference crop (RC) was defined by the Food and Agriculture Organization of the United Nations as: the vegetation with a height of 0.12 m, a surface albedo of 0.23, and r_s of 70 s m^{-1} (Allen *et al* 1998). Thus, the equation (1) yields:

$$E_p = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} U_2 \text{VPD}}{\Delta + \gamma(1 + 0.34U_2)} \quad (2)$$

where U_2 is the wind speed at a height of 2 m, and T is the air temperature. Equation (2) is known as

the PM-RC model and has been widely used in the field of hydrology, atmospheric science, and agriculture. However, in a warming climate, the value of r_s is affected by changing environmental variables such as elevated CO_2 and increased VPD, which is no longer fixed at 70 s m^{-1} (Norby *et al* 1999, Ainsworth and Rogers 2007, Medlyn *et al* 2011, Novick *et al* 2016, Liu *et al* 2020). Here we employ the optimal model proposed by Medlyn *et al* (2011) to quantify the response of leaf-level r_s (denoted as r_s^l) to external variables:

$$r_s^l = \left[g_0 + 1.6 \left(1 + \frac{g_1}{\text{VPD}^{1/2}} \right) \frac{A}{C_a} \right]^{-1} \quad (3)$$

where g_0 ($\text{mol m}^{-2} \text{ s}^{-1}$) is minimum stomatal conductance at nighttime, g_1 ($\text{kPa}^{1/2}$) is vegetation ‘marginal water use efficiency’, C_a (ppm) is the concentration of atmospheric CO_2 , and A ($\mu\text{mol m}^{-2} \text{ s}^{-1}$) is the net assimilation rate. The value of g_0 is very low and can be taken as zero (Lin *et al* 2015, Franks *et al* 2018). g_1 can be estimated by the local meteorological observations (Lin *et al* 2015). A can be calculated as $A = A_{\text{ww}} \times f(\psi_s)$, where A_{ww} ($\mu\text{mol m}^{-2} \text{ s}^{-1}$) is the net assimilation rate under non-water-stressed conditions, and $f(\psi_s)$ is the soil moisture-induced photosynthesis reduction function. The parametrizations of g_1 , A_{ww} , and $f(\psi_s)$ are given in text S1. The performance of the equation (3) has been proved against global field observations (Sabot *et al* 2022). The canopy-level r_s (denoted as r_s^c) can be further calculated as:

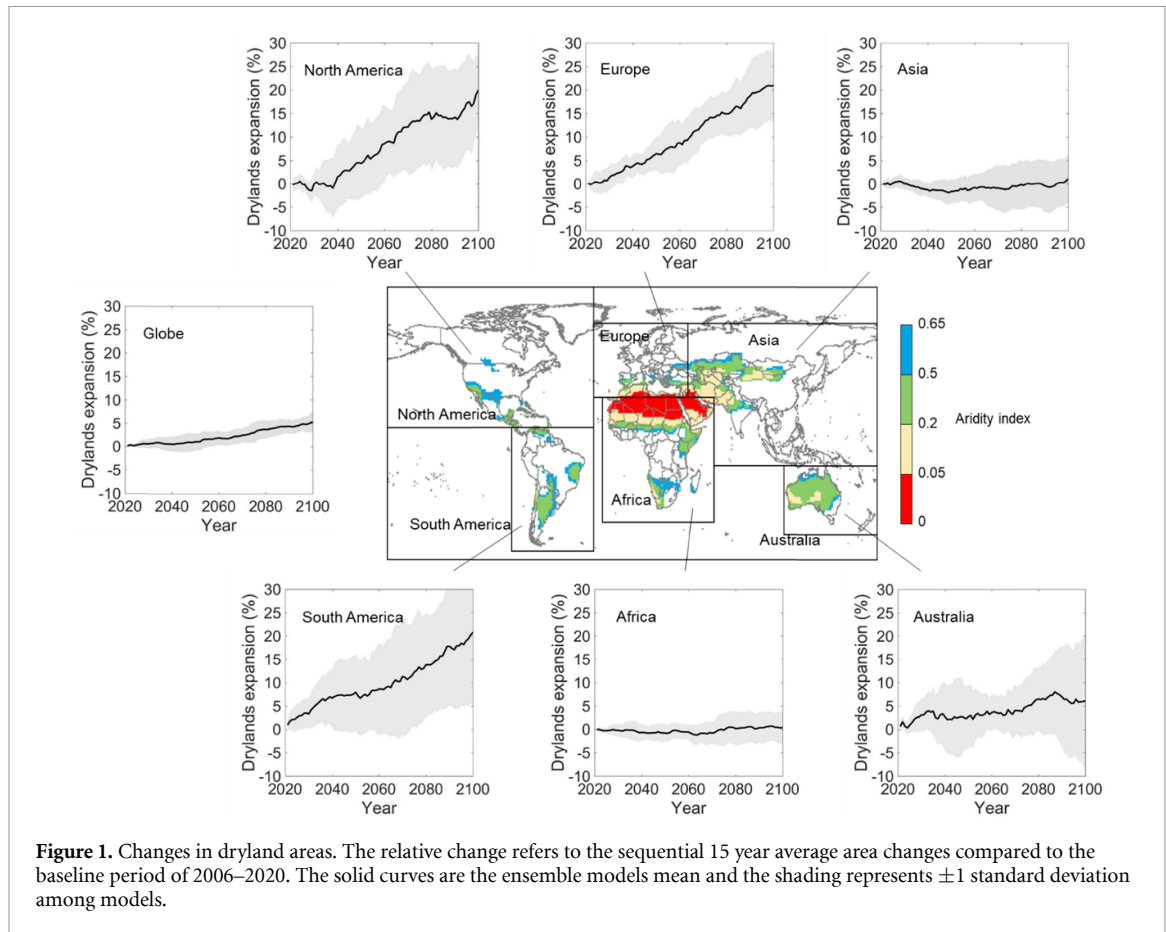
$$r_s^c = r_s^l / \text{LAI}. \quad (4)$$

Combining equations (2)–(4) and under non-water-stressed conditions (corresponding to the definition of E_p), we obtain a new E_p model (see details in the text S1):

$$E_p = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} U_2 \text{VPD}}{\Delta + \gamma \left(1 + 0.14U_2 \left(1 + \frac{g_1}{\text{VPD}^{1/2}} \right)^{-1} \frac{1}{A_{\text{ww}}} \cdot \frac{C_a}{\text{LAI}} \right)}. \quad (5)$$

Estimations of parameters g_1 and A_{ww} are shown in text S1. Equation (5) has a clear physical mechanism and allows us to fill the salient gap that E_p cannot capture the effects of rising CO_2 on vegetation.

In the following analysis, we respectively use the PM-RC model and new model to calculate AI, and detect the trends of AI changes in the coming century according to climate model data (see description below). When the vegetation response is taken into account, the AI can be used to predict potential changes in dryland areas, inherently coordinating with other indicators, such as the ecohydrologic index which also emphasizes the impacts of vegetation (Berg and McColl 2021). Drylands are divided into four subtypes: regions with $\text{AI} < 0.05$ are classified as hyperarid, 0.05–0.2 as arid, 0.2–0.5 as



semi-arid, and 0.5–0.65 as dry subhumid (Middleton and Thomas 1997, Yao *et al* 2020).

2.2. Changes in GPP

The ecosystem/canopy-level GPP and LAI are linked through assimilation per leaf area (A):

$$\text{GPP} = A \times \text{LAI}. \quad (6)$$

The fractional changes in GPP can be estimated by the fractional changes in A and LAI as:

$$\frac{\Delta \text{GPP}}{\text{GPP}} = \frac{\Delta A}{A} + \frac{\Delta \text{LAI}}{\text{LAI}} \quad (7)$$

that is, relative changes in GPP ($\Delta \text{GPP}/\text{GPP}$) can be decomposed into A -change-induced and LAI-change-induced parts. There is no direct output of A from the CMIP5 model (see description of data use below), thus the A in equation (7) is inversed by GPP/LAI .

Besides, we modify a minimalistic model to estimate relative changes in GPP as (see derivation in text S2):

$$\frac{\Delta \text{GPP}}{\text{GPP}} \approx \left(1 - \frac{\Delta \text{LAI}}{\text{LAI}}\right) \left(\frac{1 + \frac{\Delta C_a}{C_a}}{\sqrt{1 + \frac{\Delta \text{VPD}}{\text{VPD}}}} - 1\right). \quad (8)$$

It should be noted that the variables on the right-hand of equation (8) are not independent, specifically, LAI is impacted by CO_2 and VPD. In this case, the

negative sign ahead of the LAI term does not indicate a negative relationship between GPP and LAI. Equation (8) is not suitable for contribution analysis but can help us to understand the relative role of C_a and VPD on GPP changes.

2.3. Data

We use monthly outputs of 14 CMIP5 models under a high-emission scenario (RCP8.5, 2006–2100) (table S1). All models provide precipitation, surface radiations, air temperature, wind speed, air specific humidity, and LAI data, but only ten models provide GPP data. The original data are resampled to a common 1° spatial resolution by using the first-order conservative remapping scheme (Jones 1999). We calculate all variables from the individual CMIP5 model and show the standard deviation among all 14 models. To determine the parameters in the E_p model, we also select the free air CO_2 enrichment experiments data from previous studies (Ainsworth and Rogers 2007) (text S1).

3. Results

With the new E_p model, by the end of this century, we found an increase of $\sim 5\%$ (with an absolute value of 39.1% of the earth's land surface to 41.2%) of global dryland areas (figure 1), which is much smaller than the projections by PM-RC model (figure S1, with an

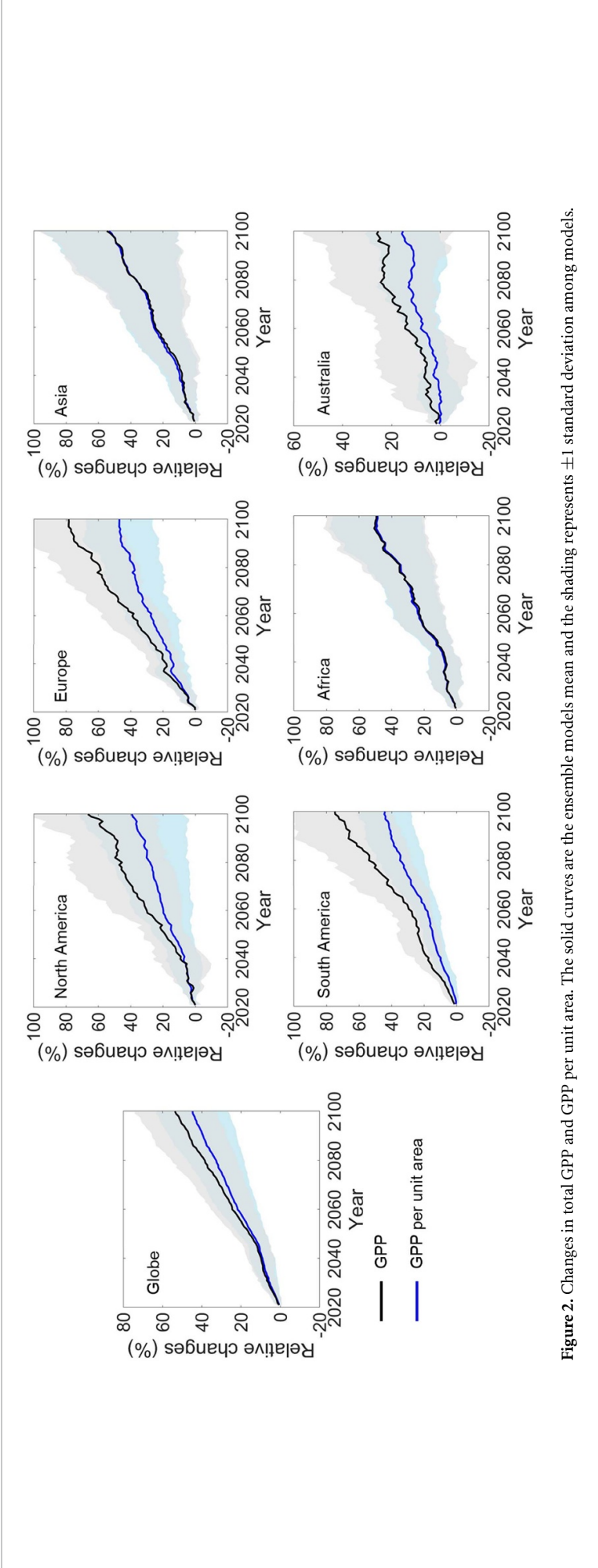


Figure 2. Changes in total GPP and GPP per unit area. The solid curves are the ensemble models mean and the shading represents ± 1 standard deviation among models.

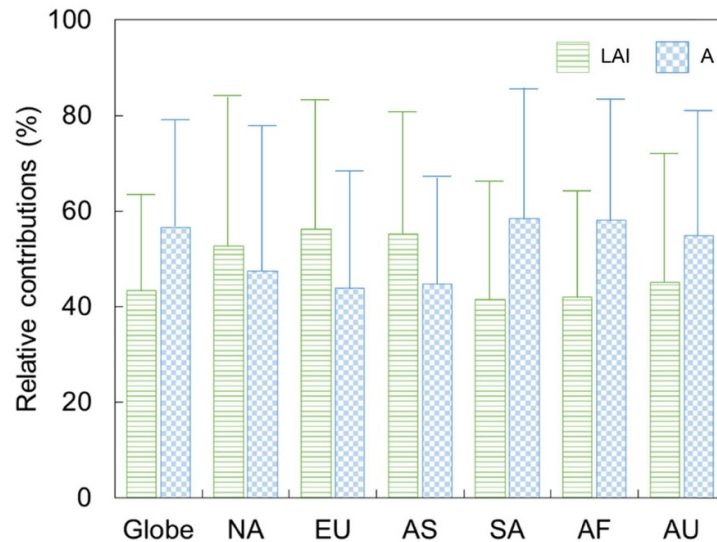


Figure 3. Contributions of LAI and assimilation (A) to GPP changes during 2086–2100 compared to 2006–2020. NA: North America, EU: Europe, AS: Asia, SA: South America, AF: Africa, and AU: Australia. The error bar represents +1 standard deviation among models.

increase of $\sim 15\%$) because much small increases in E_p (figure S2). North and South America and Europe show 15%–20% expansions, but little change is detected in Asia and Africa. This indicates that, although present dryland areas in America and Europe are not as large as Africa or Australia, they will expand much faster in relative terms in the future. Figure S3 shows the drylands expansion in four subtypes. Hyperarid drylands show the largest increase of $\sim 10\%$, the arid region increases $\sim 6\%$, and semiarid and dry sub-humid have similar increases of $\sim 4\%$. Overall, the expansion of drylands is not as severe as claimed by previous reports (which are also on the basis of AI).

The vegetation productivity shows a remarkable increase of $\sim 55\%$ across global drylands. Europe and South Africa show the largest increases of $\sim 80\%$, Australia has the lowest increase of $\sim 30\%$, and increases in other continents are $\sim 50\%$ – 60% (figure 2). GPP per unit area also indicates a significant positive trend, and the magnitudes of increases are $\sim 40\%$ – 50% among continents except for a relatively small increase of $\sim 20\%$ in Australia. GPP changes in four subtypical drylands are shown in figure S4. GPP will begin to increase after the 2050s in hyperarid drylands, while the other three subtypes show increasing trends in the near future. This may be due to the effects of CO_2 fertilization in the severe aridity regions emerge only at a high CO_2 concentration, at which CO_2 outweighs the constraints of available water.

LAI is expected to increase by $\sim 25\%$, and A shows an increase of $\sim 30\%$ by 2100 (figure S5). The decomposition analysis shows that the changes in GPP during 2086–2100 compared to the baseline of 2006–2020 are driven by concurrent increases in LAI and A (figure 3). A has an overall larger contribution across

the globe, while LAI plays a more important role in North America, Europe, and Asia. In four subtypical drylands, GPP changes in arid regions are mainly driven by increases in A, while LAI and A are comparable in other areas (figure S6).

GPP changes estimated by equation (8) is roughly consistent with direct outputs from CMIP5 models (figure S7). In 2081–2100, the relative changes in LAI, CO_2 , and VPD are $\sim 23\%$, $\sim 110\%$, and $\sim 40\%$, respectively. This suggests that despite the present and future water limits (widespread increase in atmospheric water demand), increased CO_2 will greatly enhance vegetation growth, at least until the end of this century.

4. Discussion and conclusion

The main conclusion of this study is: in a warmer climate, the expansion of global drylands is slight, but the increases in vegetation productivity are large (figures 1 and 2). Our findings contradict some existing studies, which claimed dramatic increases in global dryland areas (Dai 2013, Feng and Fu 2013, Huang *et al* 2016). The reason for the contradiction is that traditional E_p models do not consider the responses of vegetation to climate change, especially the effects of CO_2 . Those models may be suitable for historical conditions, under which the CO_2 concentration is relatively steady with slight increases. However, under future scenarios, a rapidly rising CO_2 concentration will significantly impact vegetation behavior through the leaf-level stoma and canopy structure (Ainsworth and Rogers 2007, Vicente-Serrano *et al* 2020). In the new E_p equation (equation (5)), the CO_2 effects are involved in the

leaf-level r_s^l model. The r_s^l model is physically-based and also considers other influencing factors, such as VPD. Leaf-level r_s^l and canopy-level r_s^c are linked by LAI (equation (4)), allowing us to consider both impacts of stoma and canopy structure, which outperforms traditional models under a changing climate.

Both LAI and A will increase with CO_2 fertilization (figure S5), which is inconsistent with the view of Donohue *et al* (2013). They claimed that due to the constraints of available water, the changes in A of drylands vegetation are little, and the effects of CO_2 are expressed through promoting canopy development, i.e. increases in LAI. However, the field experiments and related theories have demonstrated that despite the water limits and closure of stoma, both of A and LAI would increase at a high-level CO_2 (Long *et al* 2004, Walker *et al* 2021, Manzoni *et al* 2022). Besides, we compare the global drylands average LAI between CMIP model simulations and remote sensing observations (Moderate Resolution Imaging Spectroradiometer (MODIS) C6, (Myneni *et al* 2015)) during 2006–2020. There is only a small difference (average of 1.09 of CMIP and 1.02 of MODIS C6), indicating that uncertainty of CMIP model simulated LAI would not materially influence our conclusions, despite more or less model errors exist (Anav *et al* 2013, Park and Jeong 2021).

The increases in GPP driven by CO_2 fertilization are expected, and have been proved by global observations and vegetation models even in the historical period (Norby *et al* 2005, Zhu *et al* 2016, Zhang *et al* 2020, Lian *et al* 2021, Zhan *et al* 2022). Nevertheless, Yao *et al* (2020) pointed out that GPP per unit area will decrease by applying the historical spatial AI-GPP relationship to future scenarios. The decrease may be due to the issue of space-for-time substitution and well-mixed CO_2 in the atmosphere cannot explain the spatial variability of GPP (McColl *et al* 2022). Some underlying processes, such as the increased light interception from larger LAI and higher light use efficiency in the context of elevated CO_2 , would also in turn enhance the GPP (McCarthy *et al* 2006). Thus, we believe that both the total GPP and GPP per unit area will increase in the future (figure 1), and this is further supported by the simple diagnostic model (equation (8)), which explicitly shows the overwhelming role of CO_2 fertilization.

Finally, it should be noted that the current study is subjected to some limitations. First, in the new E_p equation, the differences in carbon assimilation taken by C3 and C4 vegetation are not considered, but we categorize the vegetation into three specific species, namely, tree, shrub, and grass. Second, a linear relationship of vegetation carbon assimilation and CO_2 is used according to field measurements, and a fine and more accurate relationship needs a further investigation. Third, the LAI used to estimate E_p is not measured under potential states (non-water-limit

conditions), but the magnitude of LAI arguably has minimal impact on long-term projections of drylands expansion (figure S8). At last, in the minimalistic model, we show that with the temperature increasing, a higher VPD will impede the vegetation growth, but temperature impacts on other mechanisms like the nutrition process and hydraulic traits are not fully represented here, and remain in future study.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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Author contributions

Z Liu and H Yang designed the study. Z Liu and T Wang conducted the formal analysis and calculation. Z Liu drafted the manuscript. All authors contributed to the results discussion and manuscript writing.

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